

H Abhijith Lal

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Professional Summary

Full Stack Software Engineer with **2.6 years of experience** specializing in Java (Spring Boot), React.js, and **Generative AI Engineering**. Expertise in developing **Text-to-SQL RAG** systems and cloud orchestration services. Proven track record in managing the full SDLC—from client discovery to production—and optimizing enterprise system performance by 25% on GCP.

Technical Skills

- **AI & GenAI:** Text-to-SQL, RAG (Retrieval-Augmented Generation), LangChain, OpenAI API, Vector Databases, Schema Linking, Prompt Engineering.
- **Languages:** Java, Python, JavaScript (ES6+), SQL (PostgreSQL, MySQL), HTML5/CSS3.
- **Frameworks:** Spring Boot, Microservices, React.js, Redux, Material UI, Spring Data JPA.
- **Cloud & DevOps:** Google Cloud Platform (Certified Cloud Engineer), Containerization, GitHub Actions, GitLab CI/CD, Terraform.
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis.
- **Testing & Tools:** JUnit, Mockito, Postman, Git, DBeaver, IntelliJ IDEA, VS Code, Cursor, Antigravity.

Professional Experience

GenTech Cloud Pvt Ltd

Sep 2023 – Present

Software Engineer

- Developed RESTful microservices using Spring Boot and React.js, enhancing platform functionality for scalable enterprise environments.
- Increased system throughput by 25% through strategic database indexing and Spring Data JPA query optimization.
- Implemented JWT-based authentication protocols, ensuring secure access and data integrity for all platform users.
- Collaborated across cross-functional teams to streamline IT processes, reducing technical debt and improving delivery efficiency.
- Optimized frontend performance by 15% by migrating legacy components to React.js and improving state management.
- Assisted in backend development with Java/Spring Boot and supported CI/CD deployments in agile cloud environments.
- Facilitated version control management and contributed to debugging complex system-level issues.

Key Projects

Cloud Resource Orchestration Platform — Client: Cognizant

Software Engineer

- Engineered a high-availability **Orchestration Service** using Spring Boot to automate the lifecycle management of cloud-native resources on GCP.
- Performed **Root Cause Analysis (RCA)** for complex orchestration failures by analyzing distributed traces and system logs, significantly improving platform stability.
- Collaborated with cross-functional infrastructure teams to resolve synchronization bugs between decoupled microservices, ensuring 99.9% resource state accuracy.
- Optimized API communication using asynchronous messaging, reducing task execution latency by 20%.

AI-Powered Financial Budgeting Platform

Full-Stack AI Engineer

- **Architected the entire product from scratch**, leading the full SDLC by gathering complex client requirements and translating them into a scalable technical blueprint.
- Engineered a marquee **Text-to-SQL RAG Chatbot** using **LangChain** and **OpenAI**, enabling natural language querying of PostgreSQL databases with 90%+ accuracy.
- Designed a custom **Schema Linking** pipeline to map natural language intent to relational table structures for real-time financial reporting.

- Adopted an **Agile methodology**, conducting iterative sprint demos for stakeholders and incorporating feedback to refine the UI/UX and AI precision.
- Developed the core full-stack system using **Spring Boot and React/Redux**, reducing data retrieval latency by 35% and initial page load times by 50%.

Education

TKM Institute of Technology, Kollam <i>Bachelor of Technology in Computer Science and Engineering</i>	Graduated 2023
GHSS, Puthoor <i>Senior Secondary Education (Computer Science)</i>	Graduated 2019